CIS437 - Cloud Computing

– What is Cloud Computing? –

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Based on the original version by Professor Erik Fredericks

Overview

- What is cloud computing?
- How did we get here?
- How does cloud usage cost money?
- What are the major cloud service models?

What is cloud computing?

When you hear cloud computing, what do you think of?

Data centers

Google Euro Data Center

Google Iowa Data Center

What is cloud computing?

If the cloud still uses physical servers, what makes it different from a traditional data center?

Cloud providers

99% of my demos will most likely be with Google Cloud

- They have a *really good* education program
  - (Very easy to get credits for teaching others)

The course primarily uses Google Cloud, but concepts transfer to AWS and Azure.

[VMs] Google Cloud Compute Engine (VMs) == AWS EC2

[Serverless] Google Cloud Cloud Functions == AWS Lambda Functions

…etc.

Same concepts, different providers

CLOUD SERVICES	aws		
Virtual Servers	Instances	VM Instances	VMs
Platform-as-a-Service	Elastic Beanstalk	App Engine	Cloud Services
Serverless Computing	Lambda	Cloud Functions	Azure Functions
Docker Management	ECS	Container Engine	Container Service
Kubernetes Management	EKS	Kubernetes Engine	Kubernetes Service
Object Storage	S3	Cloud Storage	Block Blob
Archive Storage	Glacier	Coldline	Archive Storage
File Storage	EFS	ZFS / Avere	Azure Files
Global Content Delivery	CloudFront	Cloud CDN	Delivery Network
Managed Data Warehouse	Redshift	Big Query	SQL Warehouse

Note: Cloud services and product names change frequently. Focus on the concepts rather than memorizing specific product names.

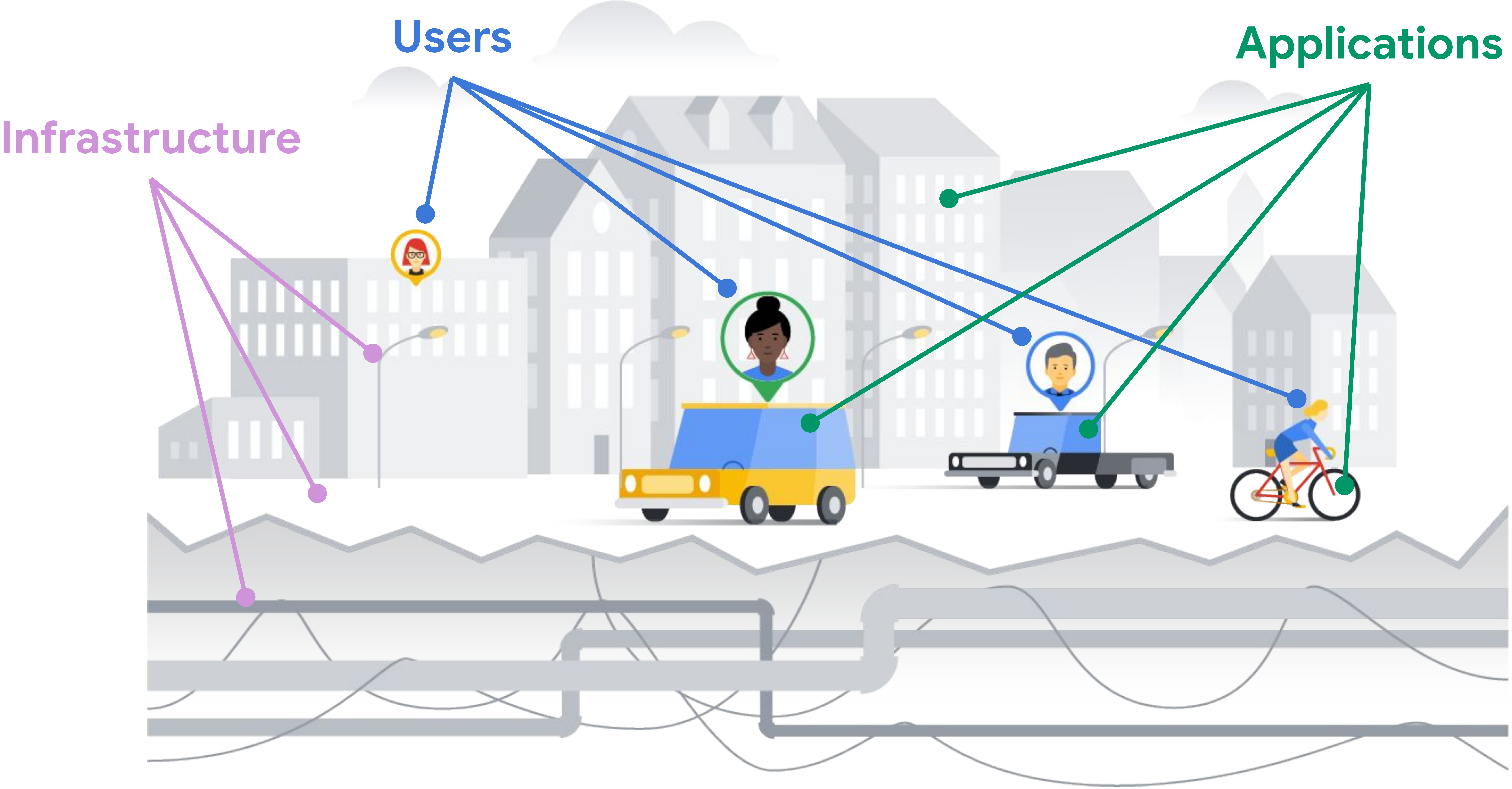
So what is cloud computing?

Cloud computing has ~~five~~ Six fundamental characteristics

On-demand self-service	Broad network access	Resource pooling	Rapid elasticity	Measured service	Unexpected costs
No human intervention needed to get resources	Access from anywhere	Provider shares resources to consumers	Get more resources quickly as needed	Pay only for what you consume	

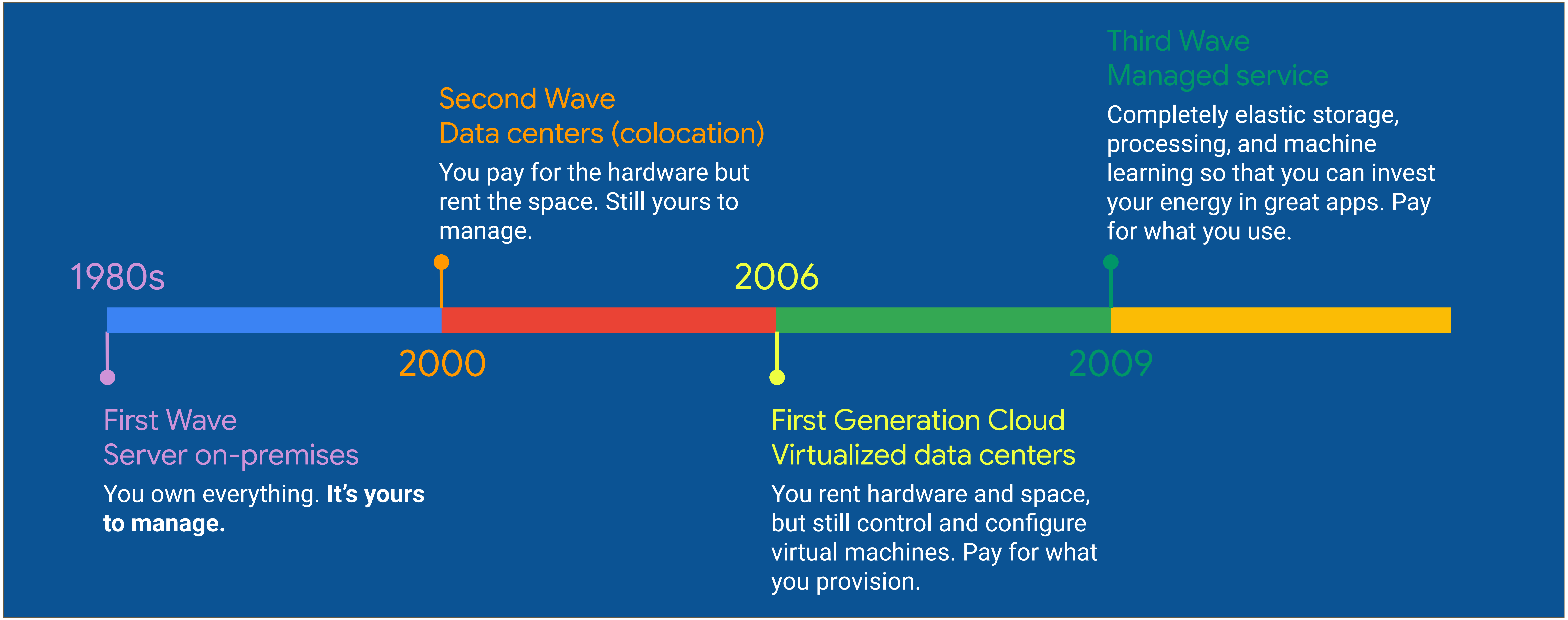
Google Cloud

An IT infrastructure is like a "city infrastructure"

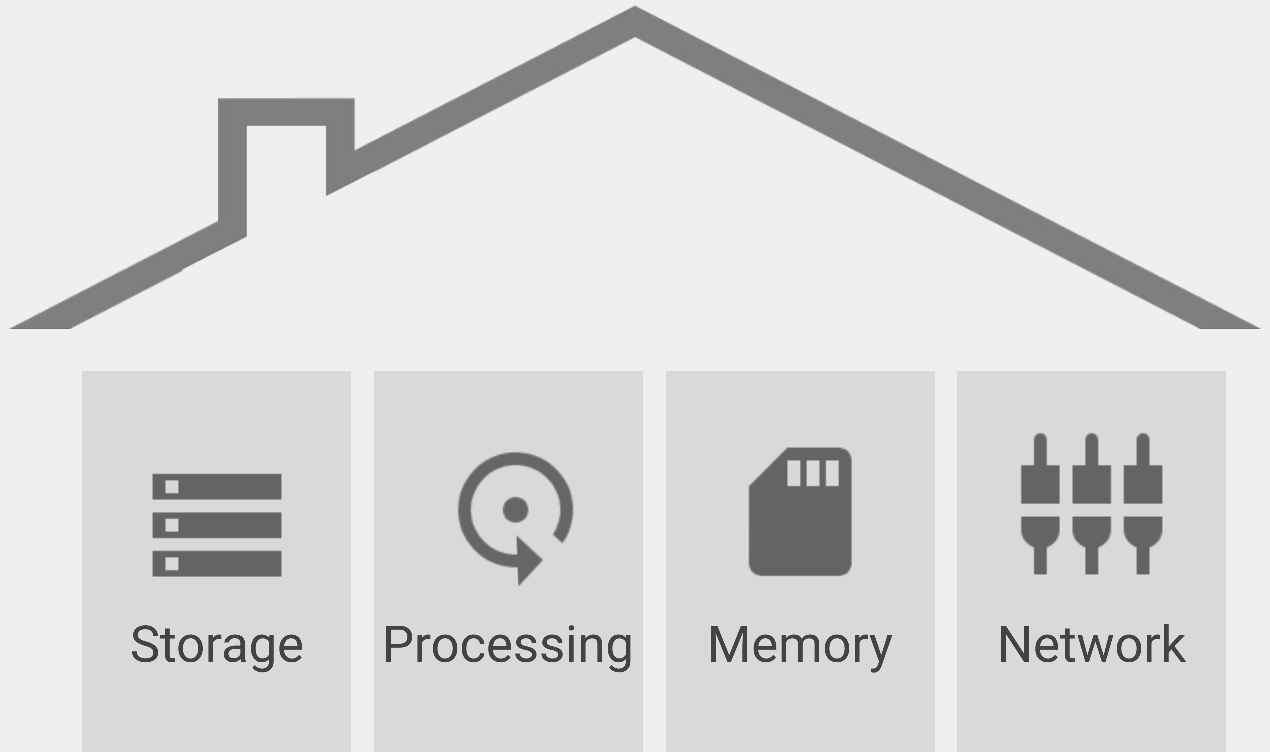


Evolution of computing infrastructure

Cloud computing is a continuation of a long-term shift in how computing resources are managed

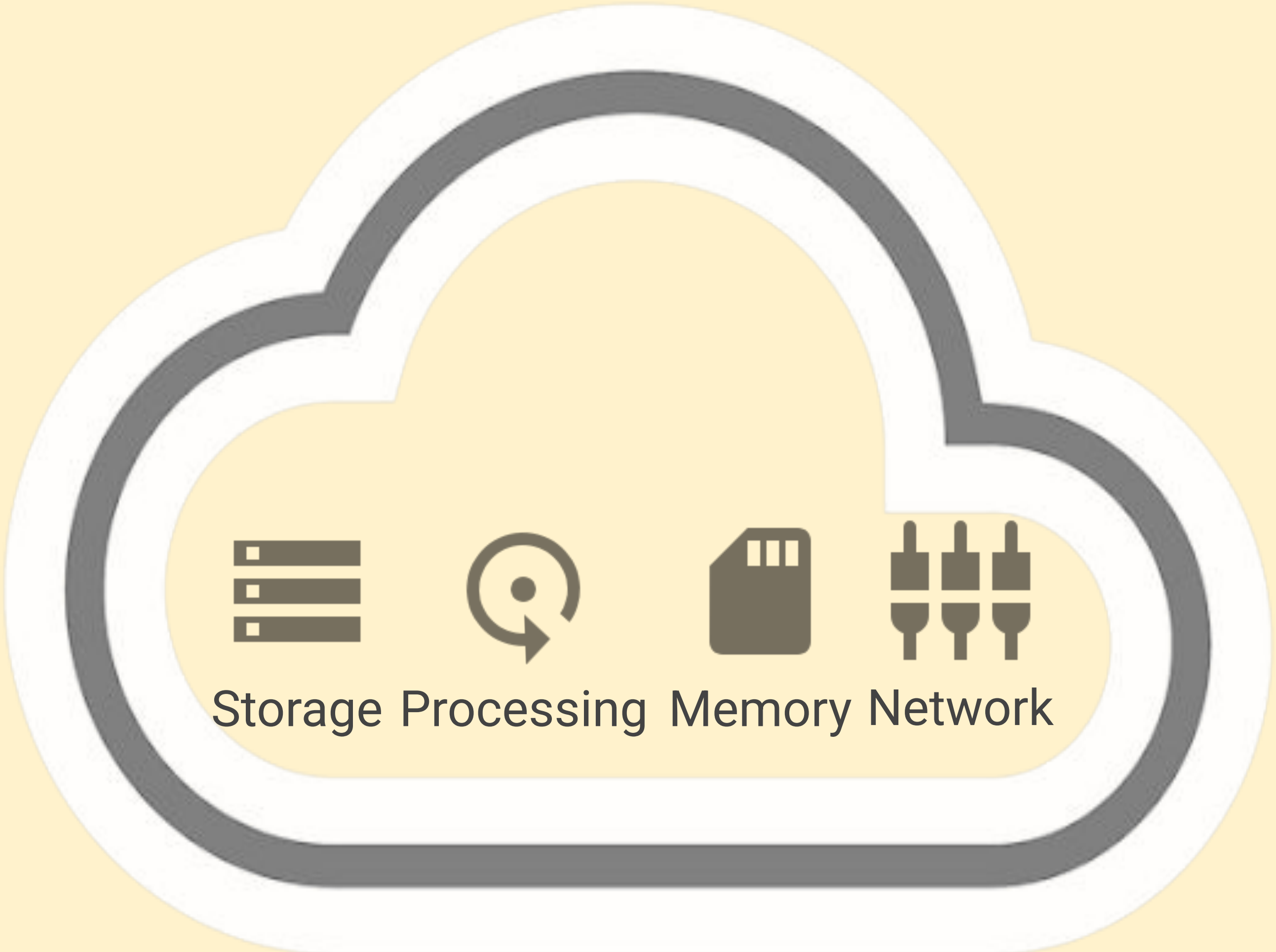


The cloud has seen a similar progression



Physical/Colocation

User-configured, managed, and maintained



Virtualized

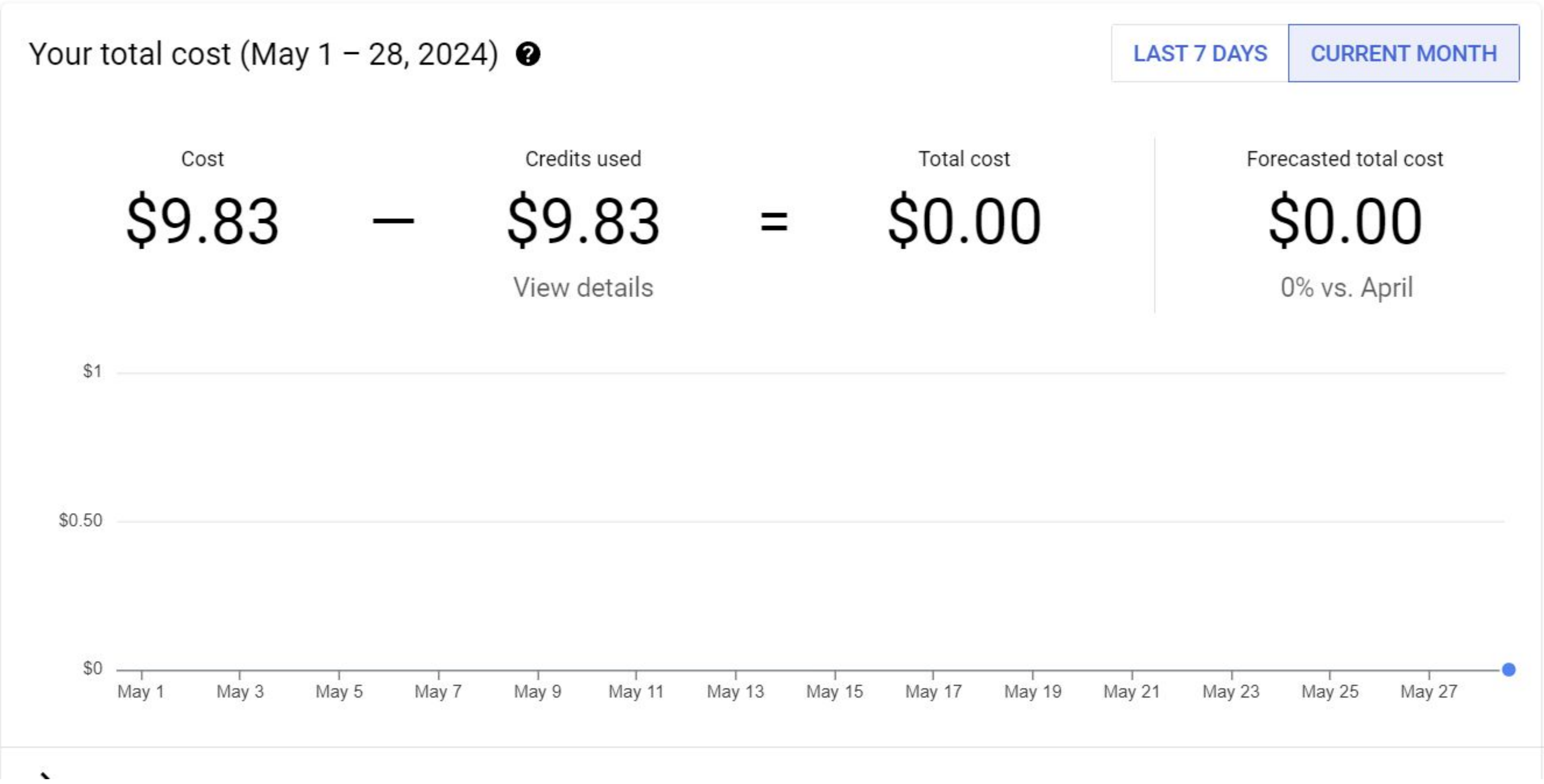
User-configured, Provider managed, and maintained

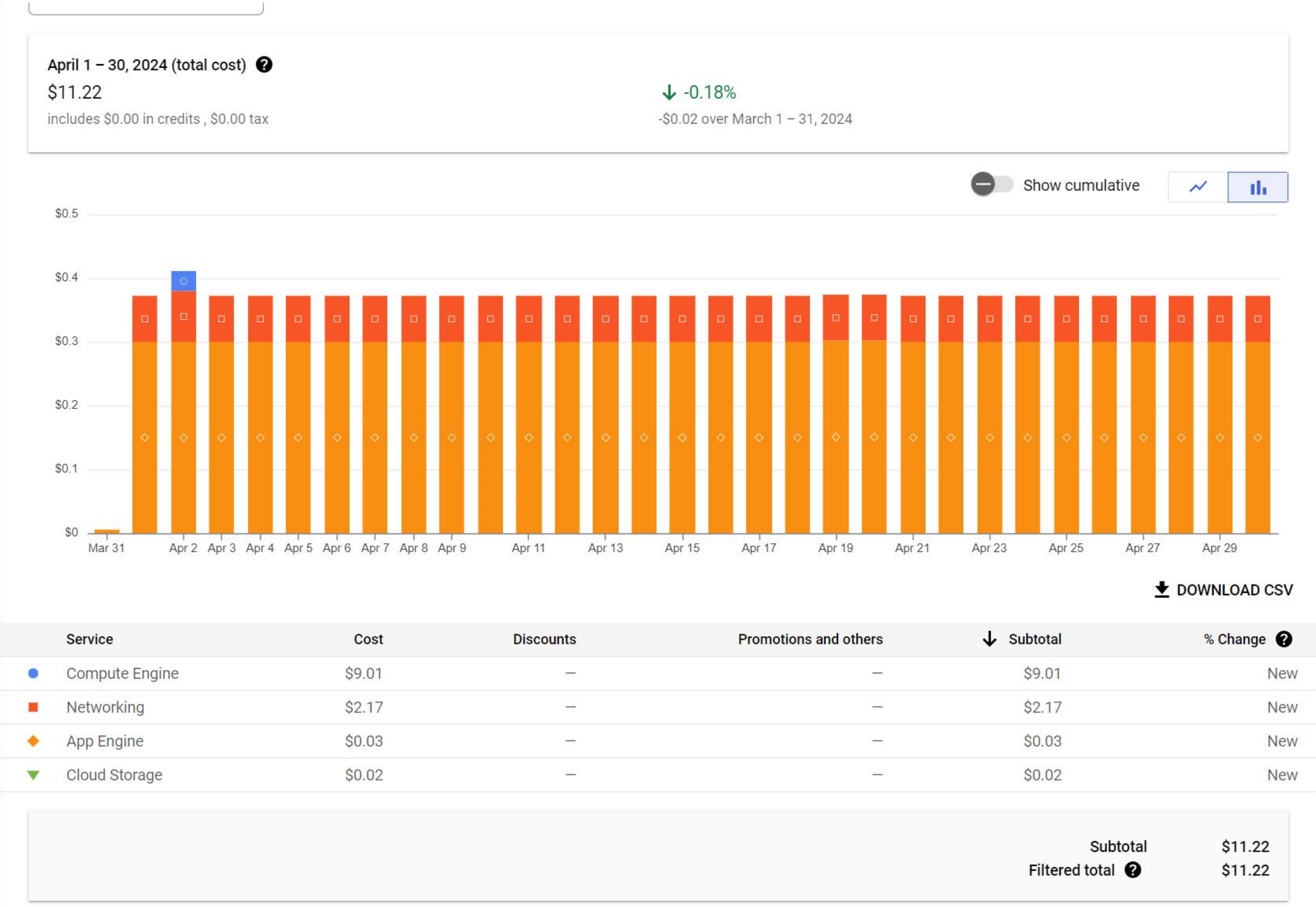
Serverless

Provider-managed, automatic provisioning/scaling.

But first…

ALWAYS CHECK YOUR BILLING STATEMENT





Unexpected cloud charges can happen.

$$ for this class

Course activities are designed to be covered by the provided educational credits.

- Do **not** put your credit card into ANY service
  - That is a great way to find auto-charges in a year when you forgot to shut things off
  -
- If you run out of credits **ask me for more**
  - I get them for free through Google

If it is on, it is costing you money

Any time *anything* is active in the cloud, you are being charged for it

- Virtual machine exists but off?
  - Charge for the hard drive storage
- Serverless function active?
  - Charge for the build/deploy activities
  - Charge for any invocations
- Look at the cloud funny?
  - Probably a charge for that too

Intention is to scare you into paying attention

Unless if otherwise specified, **turn things off when you're done**

You do not want to be out of credits the night a homework assignment is due

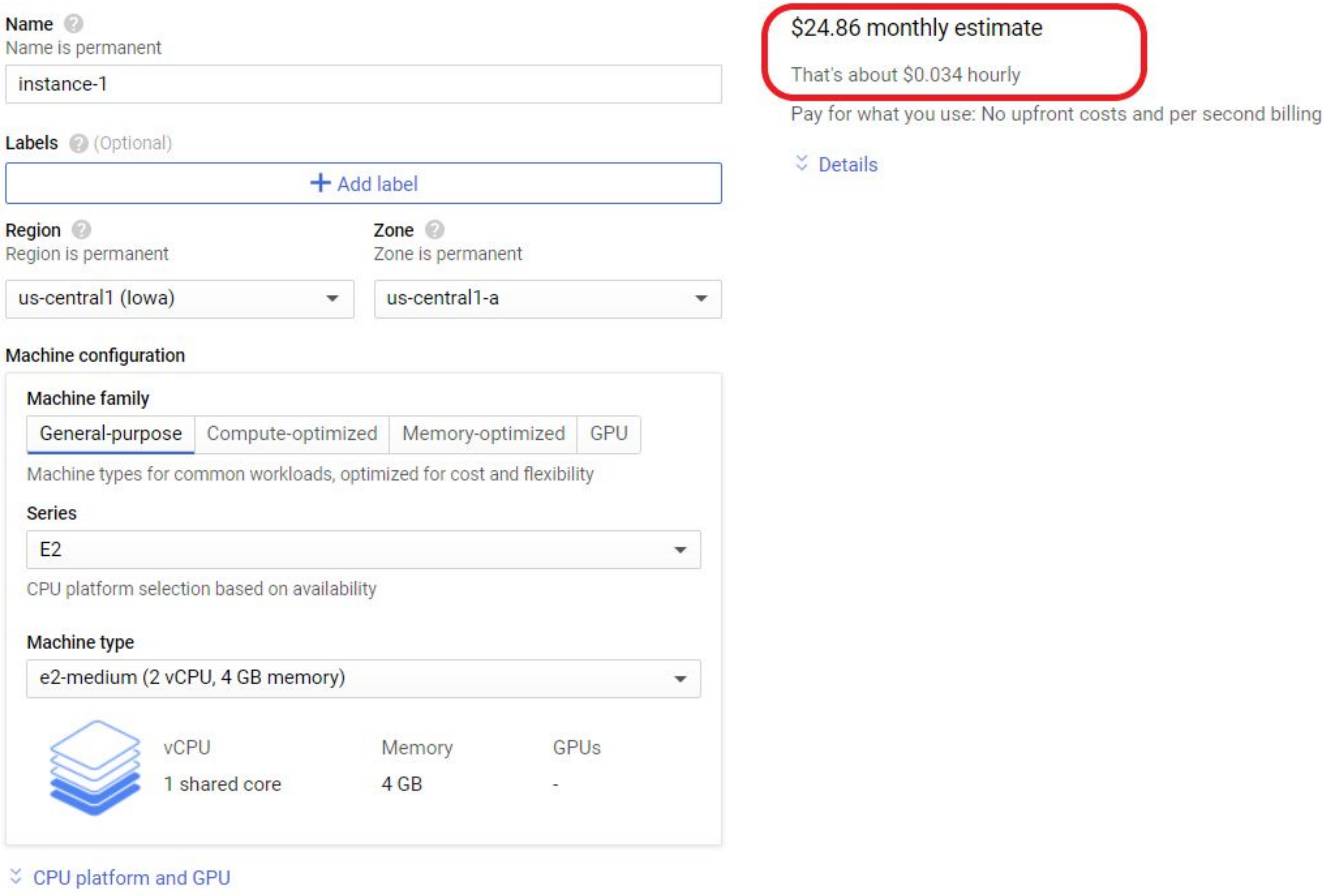
- Note: I typically don't respond to emails the evenings that assignments are due, **start early**

I can get you more credits if you let me know **ahead of time**

- Sometimes a credits request can take more time than expected

Cost and Account Safety

1. Shut down all virtual machines when not in use
2. Set up access rights for all services
3. Do not publish any keys, API information, passwords, etc. to any form of version control
4. Set quotas for all users



Quotas are helpful limits

Rate quota

GKE API: 1,000 requests per 100 seconds

Allocation quota

5 networks per project

Many quotas are changeable

How to keep your billing under control

1. Budgets and alerts
2. Billing export
3. Reports
4. Quotas

Cloud service models (IaaS / PaaS / SaaS)

Infrastructure as a service (IaaS)

- Virtualized computing infrastructure such as virtual machines, storage, and networking.
  - You manage: operating system, runtime, applications, and data
  - Example: Google Compute Engine.

Platform as a service (PaaS)

- A managed platform for deploying applications without managing the underlying servers and operating system.
  - You manage: application code and data.
  - Example: Google App Engine.

Software as a service (SaaS)

- A complete application delivered as an online service.
  - You mainly use/configure the application.
  - Example: Google Workspace / Microsoft 365.

In-class Practice: What is cloud computing? (Blackboard)